

Xcel Energy - Colorado
Monthly Gas Quality Report

– Values shown are volume weighted averages for all supplies into zone and may not represent deliveries to a specific location at a given time															
– ASTM D3588 Standard Practice for Calculating Heat Value, Compressibility Factor and Relative Density (Specific Gravity) of Gaseous Fuels and GPA 2145 Table of Physical Constants of Paraffin Hydrocarbons and Other Component of Natural Gas															
– Carbon dioxide factor based only on combustion of gas with given composition. Multiply by Gross Heating Value/1,000,000) for MT/Mscf															
– Standard conditions for gas properties are 14.73 psia, 60°F, dry, real															
ZONE	YEAR	COMPONENT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
BAYFIELD	2018	CARBON DIOXIDE	1.310	1.361	1.355	1.115	1.044	1.382	1.366	1.024	1.082				1.279
BAYFIELD	2018	OXYGEN	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0				0
BAYFIELD	2018	NITROGEN	0.528	0.695	0.820	0.528	0.640	0.522	0.551	0.646	0.618				0.639
BAYFIELD	2018	METHANE	94.059	93.111	93.177	95.253	93.254	93.197	93.460	94.505	94.149				93.729
BAYFIELD	2018	ETHANE	4.009	4.788	4.552	2.975	4.993	4.813	4.553	3.783	4.108				4.271
BAYFIELD	2018	PROPANE	0.072	0.036	0.067	0.087	0.061	0.073	0.057	0.036	0.035				0.061
BAYFIELD	2018	I-BUTANE	0.009	0.004	0.010	0.013	0.004	0.006	0.005	0.003	0.004				0.008
BAYFIELD	2018	N-BUTANE	0.009	0.005	0.011	0.017	0.004	0.006	0.006	0.003	0.004				0.009
BAYFIELD	2018	I-PENTANE	0.002	0.000	0.004	0.005	0.000	0.001	0.001	0.000	0				0.002
BAYFIELD	2018	N-PENTANE	0.001	0.000	0.002	0.003	0.000	0.000	0.000	0.000	0				0.001
BAYFIELD	2018	HEXANE-PLUS	0.001	0.000	0.002	0.004	0.000	0.000	0.001	0.000	0				0.001
BAYFIELD	2018	TOTAL	100	100	100	100	100	100	100	100	100				
BAYFIELD	2018	NET HEATING VALUE	926.8	929.5	927.7	921.8	934.7	931.7	929.5	925.7	927.9				927.7
BAYFIELD	2018	GROSS HEATING VALUE	1028.1	1031	1029	1022.9	1036.7	1033.4	1031	1027.1	1029.4				1029.1
BAYFIELD	2018	SPECIFIC GRAVITY	0.5902	0.5946	0.5945	0.5837	0.5925	0.5946	0.5932	0.5861	0.5882				0.5915
BAYFIELD	2018	WOBBE	1338.2	1337	1334.6	1338.9	1346.8	1340.2	1338.6	1341.6	1342.2				1338.1
BAYFIELD	2018	KG CO2/MMBTU	53.2	53.3	53.3	53	53.2	53.4	53.3	53	53.1				53.2
BAYFIELD	2018	COMPRESSIBILITY	0.9978	0.9978	0.9978	0.9979	0.9978	0.9978	0.9978	0.9979	0.9978				0.9978
BRUSH	2018	CARBON DIOXIDE	1.881	1.63	1.734	1.707	1.489	1.47	1.323	1.397	1.48				1.669
BRUSH	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
BRUSH	2018	NITROGEN	1.024	1.522	1.301	1.262	1.591	1.787	2.021	1.905	1.972				1.412
BRUSH	2018	METHANE	85.012	87.188	86.134	86.142	87.66	88.125	89.082	88.656	88.229				86.636
BRUSH	2018	ETHANE	10.068	7.376	8.475	8.907	7.313	6.478	5.637	6.064	6.069				8.15
BRUSH	2018	PROPANE	1.525	1.721	1.758	1.458	1.43	1.55	1.389	1.399	1.595				1.584
BRUSH	2018	I-BUTANE	0.139	0.17	0.168	0.142	0.148	0.165	0.156	0.162	0.186				0.157
BRUSH	2018	N-BUTANE	0.245	0.278	0.294	0.252	0.25	0.279	0.251	0.268	0.295				0.267
BRUSH	2018	I-PENTANE	0.041	0.048	0.052	0.048	0.047	0.057	0.054	0.058	0.065				0.049
BRUSH	2018	N-PENTANE	0.026	0.026	0.039	0.039	0.035	0.044	0.04	0.035	0.048				0.033
BRUSH	2018	HEXANE-PLUS	0.039	0.041	0.045	0.043	0.037	0.045	0.047	0.056	0.061				0.043
BRUSH	2018	TOTAL	100	100	100	100	100	100	100	100	100				
BRUSH	2018	NET HEATING VALUE	992.1	974.8	985.2	983.1	969.9	965.8	955.8	960.1	963.5				978.9
BRUSH	2018	GROSS HEATING VALUE	1097.7	1079.1	1090.4	1088	1073.9	1069.5	1058.7	1063.3	1066.9				1083.5
BRUSH	2018	SPECIFIC GRAVITY	0.6491	0.6385	0.645	0.6426	0.6336	0.6326	0.6258	0.6288	0.6331				0.6407
BRUSH	2018	WOBBE	1362.5	1350.5	1357.7	1357.2	1349.1	1344.7	1338.3	1340.9	1340.9				1353.6
BRUSH	2018	KG CO2/MMBTU	54.7	54.4	54.6	54.5	54.2	54.2	54	54.1	54.2				54.5
BRUSH	2018	COMPRESSIBILITY	0.9973	0.9975	0.9974	0.9974	0.9975	0.9975	0.9976	0.9976	0.9975				0.9974

Xcel Energy - Colorado
Monthly Gas Quality Report

– Values shown are volume weighted averages for all supplies into zone and may not represent deliveries to a specific location at a given time – ASTM D3588 Standard Practice for Calculating Heat Value, Compressibility Factor and Relative Density (Specific Gravity) of Gaseous Fuels and GPA 2145 Table of Physical Constants of Paraffin Hydrocarbons and Other Components of Natural Gas – Carbon dioxide factor based only on combustion of gas with given composition. Multiply by Gross Heating Value/1,000,000 for MT/Mscf – Standard conditions for gas properties are 14.73 psia, 60°F, dry, real															
ZONE	YEAR	COMPONENT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
DENVER	2018	CARBON DIOXIDE	1.929	1.911	1.936	1.904	1.875	1.92	1.921	2.028	2				1.927
DENVER	2018	OXYGEN	1.15	1.103	1.169	1.166	1.164	1.199	1.185	1.243	1.27				1.159
DENVER	2018	NITROGEN	4.978	4.904	5.153	5.185	5.194	5.406	5.128	5.355	5.383				5.094
DENVER	2018	METHANE	78.707	79.021	78.143	78.48	78.907	78.117	78.747	77.655	77.525				78.54
DENVER	2018	ETHANE	11.731	11.587	12.011	11.527	10.668	10.995	11.273	11.43	11.448				11.563
DENVER	2018	PROPANE	1.187	1.165	1.248	1.352	1.691	1.856	1.372	1.783	1.82				1.345
DENVER	2018	I-BUTANE	0.088	0.086	0.093	0.105	0.134	0.135	0.099	0.131	0.143				0.101
DENVER	2018	N-BUTANE	0.173	0.167	0.185	0.213	0.276	0.276	0.202	0.274	0.3				0.203
DENVER	2018	I-PENTANE	0.024	0.023	0.025	0.029	0.039	0.04	0.03	0.041	0.045				0.028
DENVER	2018	N-PENTANE	0.023	0.022	0.025	0.029	0.038	0.04	0.03	0.042	0.045				0.028
DENVER	2018	HEXANE-PLUS	0.01	0.011	0.012	0.01	0.014	0.016	0.013	0.018	0.021				0.012
DENVER	2018	TOTAL	100	100	100	100	100	100	100	100	100				
DENVER	2018	NET HEATING VALUE	947.8	947.5	949.3	948.4	949.9	952	946.8	953.2	954.7				948.9
DENVER	2018	GROSS HEATING VALUE	1048.4	1048.2	1050	1049	1050.6	1052.8	1047.3	1054	1055.6				1049.6
DENVER	2018	SPECIFIC GRAVITY	0.6741	0.6723	0.6773	0.6765	0.6771	0.6819	0.6753	0.6844	0.6856				0.6762
DENVER	2018	WOBBE	1276.9	1278.4	1275.8	1275.4	1276.8	1274.9	1274.4	1274	1274.9				1276.4
DENVER	2018	KG CO2/MMBTU	54.9	54.9	54.9	54.9	54.9	55	54.9	55.1	55.1				54.9
DENVER	2018	COMPRESSIBILITY	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974	0.9974				0.9974
GRAND JUNCTION	2018	CARBON DIOXIDE	1.792	2.036	2.133	2.455	2.496	1.802	1.67	1.403	1.549				1.967
GRAND JUNCTION	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
GRAND JUNCTION	2018	NITROGEN	0.828	0.948	1.128	1.424	1.057	0.581	0.881	1	1.084				0.984
GRAND JUNCTION	2018	METHANE	92.367	92.12	91.425	90.749	90.887	91.953	92.941	94.897	94.454				92.162
GRAND JUNCTION	2018	ETHANE	4.462	4.408	4.737	4.59	4.659	5.134	4.04	2.063	2.228				4.296
GRAND JUNCTION	2018	PROPANE	0.374	0.319	0.372	0.501	0.583	0.347	0.3	0.402	0.435				0.387
GRAND JUNCTION	2018	I-BUTANE	0.056	0.051	0.061	0.082	0.092	0.05	0.045	0.067	0.073				0.061
GRAND JUNCTION	2018	N-BUTANE	0.069	0.065	0.078	0.106	0.126	0.071	0.064	0.092	0.098				0.079
GRAND JUNCTION	2018	I-PENTANE	0.021	0.02	0.025	0.034	0.037	0.022	0.021	0.028	0.029				0.024
GRAND JUNCTION	2018	N-PENTANE	0.014	0.015	0.018	0.025	0.028	0.018	0.016	0.022	0.023				0.018
GRAND JUNCTION	2018	HEXANE-PLUS	0.017	0.018	0.023	0.034	0.035	0.022	0.022	0.026	0.027				0.022
GRAND JUNCTION	2018	TOTAL	100	100	100	100	100	100	100	100	100				
GRAND JUNCTION	2018	NET HEATING VALUE	930.8	926.2	927.7	924.7	930.2	937.7	927.3	917.5	917.4				927.5
GRAND JUNCTION	2018	GROSS HEATING VALUE	1032.2	1027.1	1028.7	1025.2	1031.2	1039.7	1028.5	1018	1017.8				1028.6
GRAND JUNCTION	2018	SPECIFIC GRAVITY	0.6038	0.6057	0.6102	0.6164	0.617	0.6061	0.6	0.5903	0.5934				0.6059
GRAND JUNCTION	2018	WOBBE	1328.4	1319.7	1316.9	1305.8	1312.8	1335.5	1327.8	1325	1321.3				1321.4
GRAND JUNCTION	2018	KG CO2/MMBTU	53.7	53.8	53.9	54.1	54.2	53.7	53.5	53.2	53.3				53.8
GRAND JUNCTION	2018	COMPRESSIBILITY	0.9978	0.9978	0.9977	0.9977	0.9977	0.9977	0.9978	0.9979	0.9979				0.9978

Xcel Energy - Colorado
Monthly Gas Quality Report

— Values shown are volume weighted averages for all supplies into zone and may not represent deliveries to a specific location at a given time															
— ASTM D3588 Standard Practice for Calculating Heat Value, Compressibility Factor and Relative Density (Specific Gravity) of Gaseous Fuels and GPA 2145 Table of Physical Constants of Paraffin Hydrocarbons and Other Component of Natural Gas															
— Carbon dioxide factor based only on combustion of gas with given composition. Multiply by Gross Heating Value/1,000,000) for MT/Mscf															
— Standard conditions for gas properties are 14.73 psia, 60°F, dry, real															
ZONE	YEAR	COMPONENT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
GREASEWOOD	2018	CARBON DIOXIDE	0.607	0.887	0.846	0.846	1.114	1.65	1.112	0.816	0.867				0.892
GREASEWOOD	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
GREASEWOOD	2018	NITROGEN	0.392	0.491	0.553	0.553	0.96	1.602	1.42	1.077	0.793				0.71
GREASEWOOD	2018	METHANE	93.605	93.286	93.115	93.115	92.077	88.644	91.84	93.791	93.239				92.869
GREASEWOOD	2018	ETHANE	5.151	4.986	5.189	5.189	5.071	6.597	4.755	3.358	4.56				5.025
GREASEWOOD	2018	PROPANE	0.167	0.26	0.219	0.219	0.576	1.191	0.669	0.646	0.341				0.366
GREASEWOOD	2018	I-BUTANE	0.025	0.033	0.028	0.028	0.074	0.128	0.081	0.111	0.07				0.05
GREASEWOOD	2018	N-BUTANE	0.028	0.039	0.039	0.039	0.084	0.143	0.092	0.141	0.088				0.061
GREASEWOOD	2018	I-PENTANE	0.009	0.01	0.007	0.007	0.02	0.025	0.017	0.031	0.02				0.013
GREASEWOOD	2018	N-PENTANE	0.006	0.006	0.004	0.004	0.013	0.014	0.01	0.021	0.017				0.009
GREASEWOOD	2018	HEXANE-PLUS	0.01	0.002	0	0	0.011	0.006	0.004	0.008	0.005				0.005
GREASEWOOD	2018	TOTAL	100	100	100	100	100	100	100	100	100				
GREASEWOOD	2018	NET HEATING VALUE	945.3	942.1	942.5	942.5	943.6	954.8	938.3	936.3	940.2				943
GREASEWOOD	2018	GROSS HEATING VALUE	1048.3	1044.8	1045.1	1045.1	1046	1057.7	1040.3	1038.3	1042.6				1045.6
GREASEWOOD	2018	SPECIFIC GRAVITY	0.5902	0.5936	0.5938	0.5938	0.603	0.626	0.6042	0.5946	0.5951				0.5965
GREASEWOOD	2018	WOBBE	1364.5	1356.1	1356.2	1356.2	1347	1336.8	1338.3	1346.5	1351.5				1353.8
GREASEWOOD	2018	KG CO2/MMBTU	53	53.2	53.2	53.2	53.4	54.1	53.4	53.1	53.2				53.2
GREASEWOOD	2018	COMPRESSIBILITY	0.9978	0.9978	0.9978	0.9978	0.9977	0.9976	0.9977	0.9978	0.9978				0.9978
MERINO	2018	CARBON DIOXIDE	0.22	0.317	0.24	0.184	0.179	0.236	0.011	0.043	0.067				0.166
MERINO	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
MERINO	2018	NITROGEN	0.256	0.284	0.276	0.244	0.242	0.255	0.228	0.237	0.234				0.251
MERINO	2018	METHANE	96.181	96.836	97.065	93.797	93.846	92.841	92.96	92.491	93.017				94.338
MERINO	2018	ETHANE	3.202	2.396	2.358	5.609	5.539	6.04	6.482	6.92	6.366				4.99
MERINO	2018	PROPANE	0.119	0.129	0.052	0.148	0.175	0.507	0.299	0.286	0.277				0.221
MERINO	2018	I-BUTANE	0.011	0.017	0.004	0.008	0.01	0.046	0.011	0.011	0.018				0.015
MERINO	2018	N-BUTANE	0.009	0.016	0.004	0.007	0.009	0.048	0.008	0.008	0.015				0.014
MERINO	2018	I-PENTANE	0.001	0.003	0	0.001	0	0.011	0	0.001	0.003				0.002
MERINO	2018	N-PENTANE	0	0.001	0	0.001	0	0.008	0	0.001	0.001				0.001
MERINO	2018	HEXANE-PLUS	0.001	0.001	0.001	0.001	0	0.008	0.001	0.002	0.002				0.002
MERINO	2018	TOTAL	100	100	100	100	100	100	100	100	100				
MERINO	2018	NET HEATING VALUE	934	927.6	926.4	952.1	952	962.1	962.3	964.9	961				949.2
MERINO	2018	GROSS HEATING VALUE	1036.4	1029.5	1028.2	1055.7	1055.6	1066.5	1066.7	1069.5	1065.3				1052.6
MERINO	2018	SPECIFIC GRAVITY	0.575	0.5725	0.5703	0.5866	0.5864	0.5944	0.5906	0.593	0.5907				0.5844
MERINO	2018	WOBBE	1366.8	1360.6	1361.5	1378.4	1378.5	1383.3	1388	1388.8	1386.1				1376.9
MERINO	2018	KG CO2/MMBTU	52.6	52.5	52.5	52.9	52.8	53.1	52.9	53	52.9				52.8
MERINO	2018	COMPRESSIBILITY	0.9979	0.9979	0.9979	0.9978	0.9978	0.9977	0.9977	0.9977	0.9977				0.9978

Xcel Energy - Colorado
Monthly Gas Quality Report

– Values shown are volume weighted averages for all supplies into zone and may not represent deliveries to a specific location at a given time															
– ASTM D3588 Standard Practice for Calculating Heat Value, Compressibility Factor and Relative Density (Specific Gravity) of Gaseous Fuels and GPA 2145 Table of Physical Constants of Paraffin Hydrocarbons and Other Component of Natural Gas															
– Carbon dioxide factor based only on combustion of gas with given composition. Multiply by Gross Heating Value/1,000,000) for MT/Mscf															
– Standard conditions for gas properties are 14.73 psia, 60°F, dry, real															
ZONE	YEAR	COMPONENT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
MOUNTAIN	2018	CARBON DIOXIDE	1.83	1.83	1.92	1.83	1.83	3.18	2.11	3.23	3.59				2.372
MOUNTAIN	2018	OXYGEN	1.37	1.26	1.2	1.21	1.44	0.61	1.05	0.54	0.37				1.006
MOUNTAIN	2018	NITROGEN	5.34	4.89	4.72	4.72	5.61	2.41	4.18	2.12	1.49				3.942
MOUNTAIN	2018	METHANE	78.77	79.01	78.7	78.9	79.11	87.94	81.2	88.49	90.26				82.487
MOUNTAIN	2018	ETHANE	12.05	11.76	12.86	12.66	10.9	4.94	9.93	4.99	3.73				9.313
MOUNTAIN	2018	PROPANE	0.51	0.96	0.5	0.54	0.88	0.74	1.3	0.58	0.46				0.719
MOUNTAIN	2018	I-BUTANE	0.04	0.08	0.03	0.04	0.06	0.05	0.07	0.02	0.03				0.047
MOUNTAIN	2018	N-BUTANE	0.07	0.16	0.05	0.08	0.13	0.1	0.12	0.03	0.05				0.088
MOUNTAIN	2018	I-PENTANE	0.01	0.02	0.01	0.01	0.02	0.02	0.02	0	0.01				0.013
MOUNTAIN	2018	N-PENTANE	0.01	0.02	0.01	0.01	0.02	0.01	0.01	0	0.01				0.011
MOUNTAIN	2018	HEXANE-PLUS	0	0.01	0	0	0	0	0.01	0	0				0.002
MOUNTAIN	2018	TOTAL	100	100	100	100	100	100	100	100	100				
MOUNTAIN	2018	NET HEATING VALUE	931.7	944.8	943.1	943.8	927.9	906.6	941	904.6	899				926.9
MOUNTAIN	2018	GROSS HEATING VALUE	1031	1045.2	1043.5	1044.2	1026.8	1005	1041.4	1002.9	997.1				1026.3
MOUNTAIN	2018	SPECIFIC GRAVITY	0.6678	0.6708	0.6686	0.6677	0.6689	0.6329	0.6629	0.6285	0.6219				0.6544
MOUNTAIN	2018	WOBBE	1261.6	1276.2	1276.2	1277.9	1255.5	1263.3	1279.1	1265	1264.4				1268.8
MOUNTAIN	2018	KG CO2/MMBTU	54.7	54.8	54.8	54.7	54.7	54.6	54.8	54.5	54.6				54.7
MOUNTAIN	2018	COMPRESSIBILITY	0.9975	0.9975	0.9975	0.9975	0.9975	0.9977	0.9975	0.9977	0.9978				0.9976
NORTHERN	2018	CARBON DIOXIDE	1.794	1.794	1.973	1.974	2.114	2.122	2.229	2.198	2.012				1.971
NORTHERN	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
NORTHERN	2018	NITROGEN	0.557	0.568	0.464	0.456	0.41	0.452	0.423	0.392	0.364				0.477
NORTHERN	2018	METHANE	86.351	86.56	85.103	85.652	84.968	84.609	84.421	83.74	84.21				85.375
NORTHERN	2018	ETHANE	9.703	9.476	10.677	10.056	10.491	10.557	10.746	11.323	11.143				10.283
NORTHERN	2018	PROPANE	1.328	1.342	1.498	1.57	1.707	1.906	1.852	2.016	1.922				1.596
NORTHERN	2018	I-BUTANE	0.065	0.062	0.071	0.082	0.087	0.093	0.088	0.103	0.113				0.08
NORTHERN	2018	N-BUTANE	0.152	0.152	0.165	0.159	0.17	0.199	0.191	0.183	0.189				0.168
NORTHERN	2018	I-PENTANE	0.021	0.019	0.021	0.02	0.022	0.023	0.022	0.02	0.023				0.021
NORTHERN	2018	N-PENTANE	0.019	0.018	0.019	0.019	0.021	0.025	0.017	0.014	0.016				0.019
NORTHERN	2018	HEXANE-PLUS	0.01	0.009	0.009	0.012	0.01	0.014	0.011	0.011	0.008				0.01
NORTHERN	2018	TOTAL	100	100	100	100	100	100	100	100	100				
NORTHERN	2018	NET HEATING VALUE	986.4	984.7	995.4	992.1	996.8	1000.6	999.9	1007	1006.6				994.1
NORTHERN	2018	GROSS HEATING VALUE	1091.8	1089.9	1101.4	1097.9	1102.9	1107	1106.1	1113.8	1113.4				1100
NORTHERN	2018	SPECIFIC GRAVITY	0.6389	0.6378	0.6469	0.6446	0.6496	0.6528	0.6537	0.6577	0.6542				0.6462
NORTHERN	2018	WOBBE	1365.9	1364.7	1369.4	1367.5	1368.4	1370.1	1368.1	1373.4	1376.6				1368.5
NORTHERN	2018	KG CO2/MMBTU	54.5	54.5	54.7	54.7	54.8	54.9	54.9	55	54.9				54.7
NORTHERN	2018	COMPRESSIBILITY	0.9974	0.9974	0.9973	0.9974	0.9973	0.9973	0.9973	0.9972	0.9973				0.9973

Xcel Energy - Colorado
Monthly Gas Quality Report

– Values shown are volume weighted averages for all supplies into zone and may not represent deliveries to a specific location at a given time – ASTM D3588 Standard Practice for Calculating Heat Value, Compressibility Factor and Relative Density (Specific Gravity) of Gaseous Fuels and GPA 2145 Table of Physical Constants of Paraffin Hydrocarbons and Other Components of Natural Gas – Carbon dioxide factor based only on combustion of gas with given composition. Multiply by Gross Heating Value/1,000,000) for MT/Mscf – Standard conditions for gas properties are 14.73 psia, 60°F, dry, real															
ZONE	YEAR	COMPONENT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
PUEBLO	2018	CARBON DIOXIDE	0.608	0.68	1.115	1.315	1.659	1.92	1.9	1.993	1.918				1.063
PUEBLO	2018	OXYGEN	0.747	0.754	0.971	1.11	1.306	1.521	1.471	1.523	1.504				0.982
PUEBLO	2018	NITROGEN	3.502	3.647	4.227	4.581	5.39	5.995	5.79	6.034	6.045				4.288
PUEBLO	2018	METHANE	94.642	93.742	88.053	85.184	79.539	76.471	77.084	75.896	76.031				88.443
PUEBLO	2018	ETHANE	0.398	0.945	4.572	6.323	9.531	11.143	11.063	11.637	11.548				4.183
PUEBLO	2018	PROPANE	0.078	0.179	0.828	1.142	1.939	2.225	2.024	2.188	2.22				0.791
PUEBLO	2018	I-BUTANE	0.008	0.016	0.065	0.093	0.166	0.187	0.17	0.182	0.184				0.066
PUEBLO	2018	N-BUTANE	0.013	0.028	0.13	0.191	0.351	0.398	0.365	0.396	0.402				0.137
PUEBLO	2018	I-PENTANE	0.002	0.004	0.017	0.026	0.05	0.058	0.054	0.06	0.06				0.02
PUEBLO	2018	N-PENTANE	0.001	0.003	0.016	0.025	0.049	0.058	0.055	0.061	0.061				0.019
PUEBLO	2018	HEXANE-PLUS	0.001	0.002	0.006	0.01	0.02	0.024	0.024	0.03	0.027				0.008
PUEBLO	2018	TOTAL	100	100	100	100	100	100	100	100	100				
PUEBLO	2018	NET HEATING VALUE	873.4	877.4	905.3	918.5	947.1	955	952.8	957.2	957.8				902.2
PUEBLO	2018	GROSS HEATING VALUE	969.8	974	1003.6	1017.4	1047.6	1055.7	1053.4	1057.9	1058.7				1000.2
PUEBLO	2018	SPECIFIC GRAVITY	0.5822	0.5877	0.6223	0.6398	0.6756	0.694	0.6894	0.697	0.6962				0.6201
PUEBLO	2018	WOBBE	1271	1270.5	1272.2	1271.9	1274.5	1267.2	1268.7	1267.2	1268.8				1270.9
PUEBLO	2018	KG CO2/MMBTU	52.4	52.6	53.5	54	54.8	55.2	55.1	55.3	55.3				53.4
PUEBLO	2018	COMPRESSIBILITY	0.9981	0.998	0.9978	0.9977	0.9975	0.9973	0.9974	0.9973	0.9973				0.9978
RIFLE-VAIL	2018	CARBON DIOXIDE	1.305	1.48	1.852	0.15	0.663	0.497	0.407	0.587	0.86				0.972
RIFLE-VAIL	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
RIFLE-VAIL	2018	NITROGEN	0.075	0.078	0.092	0.071	0.102	0.097	0.089	0.113	0.165				0.092
RIFLE-VAIL	2018	METHANE	92.633	93.145	91.679	96.27	95.695	95.707	96.138	96.135	95.246				94.342
RIFLE-VAIL	2018	ETHANE	5.103	4.337	5.142	3.28	3.174	3.432	3.007	2.757	3.47				3.961
RIFLE-VAIL	2018	PROPANE	0.633	0.688	0.882	0.168	0.268	0.202	0.26	0.297	0.194				0.456
RIFLE-VAIL	2018	I-BUTANE	0.114	0.123	0.16	0.028	0.044	0.03	0.044	0.047	0.029				0.08
RIFLE-VAIL	2018	N-BUTANE	0.094	0.103	0.134	0.022	0.035	0.023	0.036	0.044	0.024				0.066
RIFLE-VAIL	2018	I-PENTANE	0.029	0.031	0.041	0.008	0.012	0.008	0.012	0.013	0.008				0.021
RIFLE-VAIL	2018	N-PENTANE	0.002	0.002	0.002	0	0.001	0	0.001	0.001	0				0.001
RIFLE-VAIL	2018	HEXANE-PLUS	0.012	0.013	0.016	0.003	0.006	0.004	0.006	0.006	0.004				0.009
RIFLE-VAIL	2018	TOTAL	100	100	100	100	100	100	100	100	100				
RIFLE-VAIL	2018	NET HEATING VALUE	952	946.1	953	938.5	935.1	936.9	936.2	933.4	933.1				942.5
RIFLE-VAIL	2018	GROSS HEATING VALUE	1055.3	1049	1056.2	1041.3	1037.5	1039.4	1038.7	1035.6	1035.2				1045.2
RIFLE-VAIL	2018	SPECIFIC GRAVITY	0.6026	0.6015	0.6123	0.575	0.5812	0.5797	0.5778	0.5789	0.5836				0.591
RIFLE-VAIL	2018	WOBBE	1359.4	1352.6	1349.8	1373.2	1360.9	1365.2	1366.5	1361.1	1355.1				1359.7
RIFLE-VAIL	2018	KG CO2/MMBTU	53.6	53.6	53.9	52.6	52.9	52.8	52.7	52.8	53				53.2
RIFLE-VAIL	2018	COMPRESSIBILITY	0.9977	0.9977	0.9976	0.9979	0.9978	0.9978	0.9979	0.9979	0.9978				0.9978

Xcel Energy - Colorado
Monthly Gas Quality Report

– Values shown are volume weighted averages for all supplies into zone and may not represent deliveries to a specific location at a given time															
– ASTM D3588 Standard Practice for Calculating Heat Value, Compressibility Factor and Relative Density (Specific Gravity) of Gaseous Fuels and GPA 2145 Table of Physical Constants of Paraffin Hydrocarbons and Other Component of Natural Gas															
– Carbon dioxide factor based only on combustion of gas with given composition. Multiply by Gross Heating Value/1,000,000) for MT/Mscf															
– Standard conditions for gas properties are 14.73 psia, 60°F, dry, real															
ZONE	YEAR	COMPONENT	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	AVG
SOUTHERN	2018	CARBON DIOXIDE	4.491	4.406	4.323	4.37	4.318	4.503	4.513	4.535	4.521				4.437
SOUTHERN	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
SOUTHERN	2018	NITROGEN	0.031	0.03	0.033	0.035	0.038	0.034	0.035	0.035	0.036				0.033
SOUTHERN	2018	METHANE	95.238	95.327	95.406	95.357	95.406	95.224	95.212	95.19	95.201				95.291
SOUTHERN	2018	ETHANE	0.232	0.229	0.23	0.23	0.23	0.231	0.232	0.232	0.234				0.231
SOUTHERN	2018	PROPANE	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007				0.007
SOUTHERN	2018	I-BUTANE	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001				0.001
SOUTHERN	2018	N-BUTANE	0	0	0	0	0	0	0	0	0				0
SOUTHERN	2018	I-PENTANE	0	0	0	0	0	0	0	0	0				0
SOUTHERN	2018	N-PENTANE	0	0	0	0	0	0	0	0	0				0
SOUTHERN	2018	HEXANE-PLUS	0	0	0	0	0	0	0	0	0				0
SOUTHERN	2018	TOTAL	100	100	100	100	100	100	100	100	100				
SOUTHERN	2018	NET HEATING VALUE	873.9	874.6	875.4	874.9	875.4	873.7	873.6	873.4	873.6				874.3
SOUTHERN	2018	GROSS HEATING VALUE	970.5	971.3	972.2	971.7	972.2	970.3	970.2	970	970.2				971
SOUTHERN	2018	SPECIFIC GRAVITY	0.5995	0.5987	0.5979	0.5984	0.5979	0.5997	0.5998	0.6	0.5999				0.599
SOUTHERN	2018	WOBBE	1253.4	1255.3	1257.3	1256.1	1257.3	1253	1252.7	1252.3	1252.6				1254.6
SOUTHERN	2018	KG CO2/MMBTU	54.5	54.5	54.4	54.4	54.4	54.5	54.5	54.5	54.5				54.5
SOUTHERN	2018	COMPRESSIBILITY	0.9979	0.9979	0.9979	0.9979	0.9979	0.9979	0.9979	0.9979	0.9979				0.9979
STERLING	2018	CARBON DIOXIDE	0.482	0.478	0.511	0.53	0.526	0.525	0.529	0.527	0.497				0.496
STERLING	2018	OXYGEN	0	0	0	0	0	0	0	0	0				0
STERLING	2018	NITROGEN	3.589	3.617	3.726	3.644	3.738	3.737	3.678	3.646	3.781				3.639
STERLING	2018	METHANE	93.492	93.591	93.841	93.84	93.823	93.826	93.716	93.599	93.79				93.652
STERLING	2018	ETHANE	1.53	1.426	1.102	1.155	1.077	1.075	1.232	1.365	1.103				1.343
STERLING	2018	PROPANE	0.571	0.551	0.476	0.485	0.474	0.475	0.491	0.516	0.482				0.53
STERLING	2018	I-BUTANE	0.101	0.101	0.103	0.103	0.105	0.105	0.101	0.101	0.103				0.102
STERLING	2018	N-BUTANE	0.123	0.119	0.118	0.119	0.122	0.122	0.12	0.119	0.119				0.12
STERLING	2018	I-PENTANE	0.039	0.039	0.042	0.042	0.044	0.044	0.041	0.041	0.041				0.04
STERLING	2018	N-PENTANE	0.024	0.025	0.026	0.026	0.028	0.027	0.026	0.026	0.026				0.025
STERLING	2018	HEXANE-PLUS	0.049	0.053	0.055	0.056	0.063	0.064	0.066	0.06	0.058				0.053
STERLING	2018	TOTAL	100	100	100	100	100	100	100	100	100				
STERLING	2018	NET HEATING VALUE	903.4	902.2	897.7	898.8	897.7	897.8	899.4	900.8	897.5				901
STERLING	2018	GROSS HEATING VALUE	1002.3	1001	996.1	997.3	996.2	996.2	998	999.5	995.9				999.7
STERLING	2018	SPECIFIC GRAVITY	0.5929	0.5924	0.5909	0.5911	0.5913	0.5913	0.5919	0.5925	0.5911				0.592
STERLING	2018	WOBBE	1301.7	1300.5	1295.8	1297.2	1295.5	1295.5	1297.2	1298.5	1295.3				1299.3
STERLING	2018	KG CO2/MMBTU	52.8	52.8	52.7	52.8	52.7	52.7	52.8	52.8	52.7				52.8
STERLING	2018	COMPRESSIBILITY	0.9979	0.9979	0.998	0.998	0.998	0.998	0.998	0.998	0.998				0.9979